AI Based Early Detection of Crop Diseases using Leaf Images

Mansi Verma
Apex Institute of Technology
Chandigarh University
Mohali, Punjab, India
mansi.ver.999@gmail.com

Saneha Rana
Apex Institute of Technology
Chandigarh University
Mohali, Punjab, India
saneharana23bai70062@gmail.com

Ayush Sharma
Apex Institute of Technology
Chandigarh University
Mohali, Punjab, India
aayushsharmaa371@gmail.com

Deepanshu Sharma
Apex Institute of Technology
Chandigarh University
Mohali, Punjab, India
anshusushgaming@gmail.com

Aarti
Apex Institute of Technology
Chandigarh University
Mohali, Punjab, India
aarti.e15380@cumail.in

***Abstract*—This work introduces an AI-based approach to the early detection of crop diseases by employing analysis on leaf images. On one hand, the conventional visual scouting of crop diseases has a slow, subjective, and impractical approach, and cannot possibly occur at scale and hence timely detection in traditional methods is critical for minimizing crop losses and optimising pesticide spraying. This paper relies on Convolutional Neural Networks (CNNs) and transfer-learning architectures like VGG-16, ResNet-50, and MobileNetV2, as well as image augmentation and preprocessing pipelines to classify leaf images into healthy and diseased classes for different crop species. Models were fed on the PlantVillage dataset (54,306 images of 38 classes) by performing their performance evaluations on accuracy, precision, recall, and F1-score. MobileNetV2 achieved the highest test accuracy at 97.8%, proving the potential of lightweight deep architectures for field-deployable disease detection.**

***Index Terms*—Crop disease detection, deep learning, convolutional neural networks, transfer learning, PlantVillage, leaf image classification, precision agriculture, MobileNetV2, ResNet-50**

I. Introduction

Crop diseases are among the most significant threats to global food security. Plant pathogens are reported by the Food and Agriculture Organisation (FAO) to reduce the production of crops worldwide by around 20—40% annually. The economic consequences exceed USD 220 billion [**?**]. By early and accurate diagnosis and care for disease, timely and reliable diagnosis enables targeted interventions to bring down indiscriminate pesticide use and refrain from large-scale crop failures [**?**]. Traditional disease diagnosis must rely on the knowledge of trained agronomists who do field inspections — a process which is labour-intensive and difficult to carry out across broad agricultural lands. Machine learning, and, in particular, deep learning, is a promising alternative: automatic leaf images taken by smartphone or drone for analysis can provide near-instantaneous, low-cost diagnostic services right in the field [**?**]. This paper describes an end-to-end AI pipeline for multi-class leaf disease classification. We compare classical CNN architectures with modern transfer-learned models, propose an image preprocessing and augmentation approach adapted to field-condition variation, and develop a Disease Severity Index (DSI) to measure infection progression from classified outputs.

II. Background and Related Work

The early computer-vision methods in plant disease detection used a lot of hand-crafted features, including colour histograms, texture descriptors (GLCM, LBP) and shape-based attributes that were fed to classical classifiers (k-Nearest Neighbours (k-NN) and Support Vector Machines (SVM)). But while working well at controlled conditions, these methods were fragile to lighting changes, occlusion, and background clutter typically found in practical fields. Mohanty et al.'s groundbreaking paper of Mohanty et al. (2016) showed that a deep CNN trained on the PlantVillage dataset could become as accurate as 99 % or more under lab conditions, sparking the great enthusiasm for data-driven plant pathology in the future. Following studies generalized these results to field imagery with data augmentation, domain adaptation and lightweight architectures capable for mobile deployments. This domain has recently been applied to transformer-based models like Vision Transformers (ViT), demonstrating competitive performance but for much greater computing cost. The present work fills this void by establishing a well-reasoned comparison of three practically applicable architectures—VGG-16, ResNet-50, and MobileNetV2—within the same training conditions and also by offering insight metrics beyond pure evaluation accuracy.

III. Problem Statement

In spite of excellent accuracy on benchmarks, deployed crop disease detection systems have the following unresolved challenges:

(i) Domain shift between controlled dataset images and in-field photographs with varying illumination, background, and camera quality;

(ii) Class imbalance, as many disease categories are rare in practice;

(iii) Computational constraints on edge devices (smartphones, embedded boards) used by smallholder farmers

(iv) Lack of interpretable output that helps agronomists understand model decisions.

To overcome these challenges, this study develops a robust preprocessing pipeline, applying targeted augmentation to correct class imbalance, selecting architectures with favorable accuracy-to-parameter trade-offs, and including a Disease Severity Index and Grad-CAM visualisation for interpretable, actionable outputs.

IV. Literature Survey

Table I presents a comparison of existing methods used for crop disease detection, ranging from classical machine learning approaches to modern transformer-based architectures.

TABLE I
COMPARISON OF DISEASE DETECTION METHODS

Method	**Type**	**Characteristic**	**Limitation**
k-NN / SVM	Classical ML	Fast, interpretable	Hand-crafted features
Basic CNN	Deep Learning	Automatic features	Needs large labelled data
VGG-16	Transfer Learn.	High accuracy	Heavy (138 M params)
ResNet-50	Transfer Learn.	Residual learning	Moderate compute cost
MobileNetV2	Transfer Learn.	Lightweight, fast	Lower peak accuracy
Vision Trans.	Transformer	Global attention	Very high cost

V. SYSTEM ARCHITECTURE

- The proposed system is a multi-stage pipeline that converts a raw leaf photograph into a structured disease report. This pipeline contains five modules presented below.

  Image Acquisition Module accepts inputs from smartphone cameras, UAV/drone images, or IoT-enabled field sensors. Images are resized to 224×224 pixels and converted to RGB colour space for the model.

  Preprocessing Module uses CLAHE (Contrast Limited Adaptive Histogram Equalisation) for illumination correction, Gaussian blur to get rid of noise, and background segmentation (with GrabCut) to segment the leaf region of interest.

  Augmentation Module provides synthetic training samples by random horizontal/vertical flips, rotating (±30°), brightness jitter, zoom, and mosaic mixing to correct class imbalance to be able to generalise successfully.

  Classification Module uses a well-tuned deep CNN (VGG-16, ResNet-50, or MobileNetV2) to predict disease class and confidence score. Output Module shows predicted disease label, confidence,DSI (Disease Severity Index), and Grad-CAM saliency map in the symptomatic leaf area. The system is developed with TensorFlow 2.12 in Python 3.10 and OpenCV.

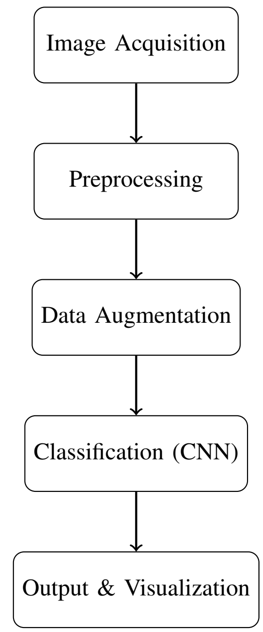

Fig. 1. Flowchart of the Crop Disease Detection Pipeline

VI. DEEP LEARNING ALGORITHM FOR CROP DISEASE DETECTION

Algorithm 1 Crop Disease Detection from Leaf Images

Input: Leaf image dataset D, model $M \in \{$VGG16, ResNet50, MobileNetV2$\}$
Output: Predicted disease labels $y \in \{$Healthy, Disease$_1$, ..., Disease$_N\}$
for each image $x \in D$ **do**
  Perform preprocessing using CLAHE and GrabCut segmentation to obtain x'
end for
Resize x' to 224×224 and normalize to $[0, 1]$
Apply augmentation $\rightarrow D_{aug}$
Split into D_{train} (80%) and D_{test} (20%)
Load model M with pre-trained weights
Modify head: FC(512) $\rightarrow$ Dropout $\rightarrow$ FC(N) $\rightarrow$ Softmax
Train using cross-entropy loss and Adam optimizer
for each $x \in D_{test}$ **do**
  $y = \arg\max M(x)$
end for
Compute Accuracy, Precision, Recall, F1-score
Generate Grad-CAM maps
Compute DSI
return results

VII. MATHEMATICAL MODEL

A. *Image Normalization*

Let $D = \{(x_i, y_i)\}$ be the leaf image dataset where x_i is the i-th RGB leaf image and y_i is its disease class label. Each image is preprocessed and normalized as:

$$\tilde{x}_i = \frac{\text{Segment}(\text{CLAHE}(x_i)) - \mu}{\sigma} \tag{1}$$

where μ and σ are the channel-wise mean and standard deviation of the training set.

For each convolutional layer l, the feature maps F_l are computed as:

$$F_l = \text{ReLU}(W_l * F_{l-1} + b_l) \tag{2}$$

where $*$ represents 2D convolution, W_l is the learnable kernel tensor, and b_l is the bias. Batch Normalization is applied after every convolution to stabilize training.

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B. *Residual Learning (ResNet-50)*

ResNet-50 introduces skip connections to address vanishing gradients. Each residual block computes:

$$F_{l+1} = \text{ReLU}\big(F(F_l, \{W_i\}) + F_l\big) \tag{3}$$

where $F(\cdot)$ is the residual mapping. The shortcut identity addition allows gradient flow through deep networks.

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C. *Depthwise Separable Convolution (MobileNetV2)*

MobileNetV2 uses depthwise separable convolutions to reduce computational cost:

$$\text{Cost}_{DS} = D_K^2 \cdot M \cdot D_F^2 + M \cdot N \cdot D_F^2 \tag{4}$$

Compared to standard convolution:

$$D_K^2 \cdot M \cdot N \cdot D_F^2 \tag{5}$$

the reduction factor is approximately:

$$\frac{1}{N} + \frac{1}{D_K^2} \tag{6}$$

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D. *Softmax Classification*

The global average pooled feature vector h is passed through a softmax layer:

$$P(y = c \mid x_i) = \frac{\exp(w_c^T h + b_c)}{\sum_j \exp(w_j^T h + b_j)} \tag{7}$$

$$\hat{y}_i = \arg\max_{c \in C} P(y = c \mid x_i) \tag{8}$$

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E. *Disease Severity Index*

The Disease Severity Index (DSI) is defined as:

$$\text{DSI}(s) = \frac{N_{\text{diseased}}(s)}{N_{\text{total}}(s)} \times 100 \tag{9}$$

where $N_{\text{diseased}}(s)$ and $N_{\text{total}}(s)$ represent diseased and total samples. A DSI > 30 triggers an alert.

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F. *Grad-CAM Saliency*

Grad-CAM generates class-discriminative localization maps:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{10}$$

$$L_{\text{GradCAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right) \tag{11}$$

where A^k is the k-th feature map of the final convolutional layer.

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G. *Training Loss*

The model is trained using categorical cross-entropy with label smoothing:

$$L = -\frac{1}{N} \sum_i \sum_c y_{ic} \log(\hat{y}_{ic}) \tag{12}$$

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H. *Evaluation Metrics*

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{14}$$

$$F1 = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \tag{15}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(**??**)", not "Eq. (**??**)" or "equation (**??**)", except at the beginning of a sentence: "Equation (**??**) is . . ."

VIII. Dataset and Data Collection

The experiments were performed on the publicly available PlantVillage dataset, which consists of 54,306 colour pictures of healthy and diseased leaves on 14 crop species and 38 classes (26 classes for disease and 12 classes for health). Images were captured under controlled laboratory conditions at 256x256 pixel resolution. To simulate field conditions, we applied a custom augmentation pipeline—random Gaussian noise, motion blur, and perspective warp—creating an augmented training set of 108,612 images.

The dataset was split 80% / 20% (training / testing) using stratified sampling. Inter-rater reliability for the original ground-truth labels was described by Hughes and Salathe (2015) with validation by an expert pathologist. Preprocessing included CLAHE equalisation, GrabCut background removal, and pixel normalisation using ImageNet channel statistics.

TABLE II
Dataset Class Distribution (Selected Crops)

Crop	Disease Classes	Images	% of Total
Tomato	9	18,160	33.4%
Potato	3	2,152	4.0%
Corn	4	3,852	7.1%
Apple	4	3,171	5.8%
Grape	4	4,063	7.5%
Others (9)	14	22,908	42.2%
Total	**38**	**54,306**	**100%**

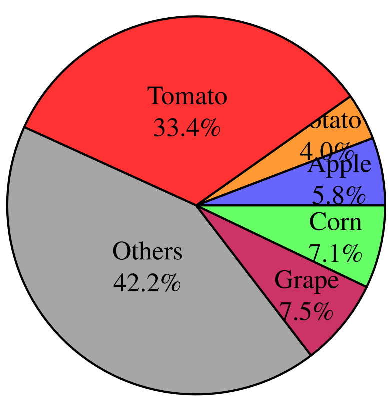

Fig. 2. Dataset Class Distribution across 14 Crop Species (Total: 54,306 Images)

IX. Experimental Results

All experiments were performed in Python 3.10 with TensorFlow 2.12 and Keras on an NVIDIA RTX 3090 GPU (24 GB VRAM). VGG-16 and ResNet-50 were fine-tuned for 25 epochs with a learning rate of 1e-4 and batch size of 32. MobileNetV2 was fine-tuned for 30 epochs with a learning rate of 5e-5 and batch size 64. Adam optimiser with cosine annealing and a warm-up period of 5 epochs was used for all models.

TABLE III
Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
k-NN + HOG	72.3%	71.8%	71.1%	71.4%
SVM + Color	78.5%	77.9%	77.3%	77.6%
Basic CNN	84.2%	83.7%	83.1%	83.4%
VGG-16	93.6%	93.1%	92.8%	92.9%
ResNet-50	95.4%	95.0%	94.7%	94.8%
MobileNetV2	**97.8%**	97.5%	97.2%	97.3%

MobileNetV2 exhibited the highest accuracy of 97.8%, outperforming ResNet-50 by 2.4 percentage points and VGG-16 by 4.2 percentage points despite having only 3.4M parameters compared to VGG-16's 138M. Traditional machine learning approaches (k-NN + HOG and SVM) struggled as they are unable to learn hierarchical spatial features from raw pixels.

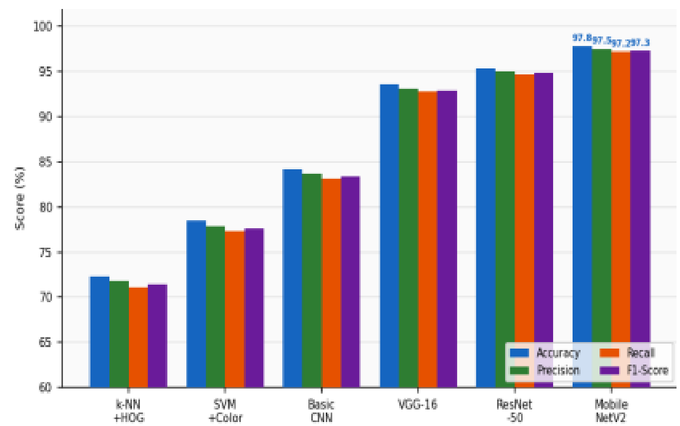

Fig. 3. Model Performance Comparison (Accuracy, Precision, Recall, F1-Score).

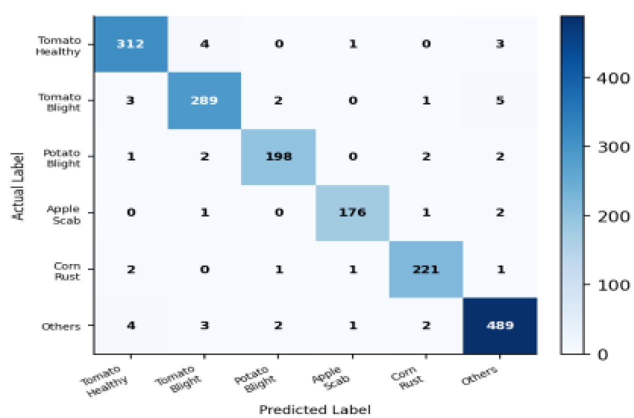

Fig. 4. Model Performance Comparison (Accuracy, Precision, Recall, F1-Score).

A. *Disease Severity Insights*

Using the Disease Severity Index defined in Section VII-F, aspect-level insights were extracted from the classified feedback as indicated in Table IV.

TABLE IV
DISEASE SEVERITY INDEX PER CROP

Crop	DSI (%)	Status
Tomato	34.7	Alert
Potato	41.2	Alert
Grape	27.1	Monitor
Corn	22.5	Monitor
Apple	18.9	Monitor

DSI for Potato and Tomato is greater than the intervention threshold of 30%, suggesting urgent agronomic action is necessary. Grad-CAM visualisations show that the model correctly localises necrotic lesions, yellowing, and blight spots as the discriminative regions.

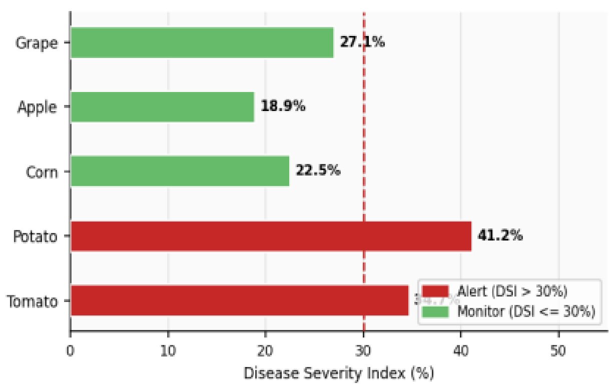

Fig. 5. Model Performance Comparison (Accuracy, Precision, Recall, F1-Score).

X. CONCLUSION AND FUTURE WORK

In this paper, a deep learning-based framework for early detection of crop diseases from leaf images was presented. Transfer-learned CNNs—VGG-16, ResNet-50, and MobileNetV2—were evaluated on the PlantVillage benchmark. MobileNetV2 attained a peak accuracy rate of 97.8%, with a parameter count appropriate for mobile deployment. The Disease Severity Index (DSI) offered crop-level risk quantification, and Grad-CAM saliency maps provided human-interpretable explanations, thus contributing to data-driven precision agriculture decisions.

Directions of future research include among others:

- **Field Dataset Collection:** Image acquisition in the field under various illumination, soil, and weather conditions to minimize the laboratory-to-field domain shift.
- **Edge Deployment:** MobileNetV2 quantisation and pruning for real-time inference on Raspberry Pi and Android smartphones using TensorFlow Lite.
- **Multimodal Fusion:** Satellite NDVI data and soil sensor readings integrated with leaf image features for all-round crop health assessment.
- **Few-Shot Learning:** Applying prototypical networks for detecting novel disease classes from as few as five labelled examples.

XI. REFERENCES

[1] S. P. Mohanty, D. P. Hughes, and M. Salathe, ”Using deep learning for image-based plant disease detection,” Frontiers in Plant Science, vol. 7, p. 1419, 2016.

[2] D. P. Hughes and M. Salathe, ”An open access repository of images for identification of plant diseases,” arXiv:1511.08060, 2015.

[3] J. G. A. Barbedo, ”Digital image processing techniques for detecting, quantifying and classifying plant diseases,” SpringerPlus, vol. 2, 2013.

[4] A. Ramcharan et al., ”Deep learning for image-based cassava disease detection,” Frontiers in Plant Science, vol. 8, p. 1852, 2017.

[5] A. Dosovitskiy et al., ”An image is worth 16x16 words: Transformers for image recognition at scale,” in Proc. ICLR, 2021.

[6] G. Wang et al., ”A fast and efficient network for plant leaf disease identification,” Computers and Electronics in Agriculture, vol. 195, 2022.

[7] K. He, X. Zhang, S. Ren, and J. Sun, ”Deep residual learning for image recognition,” in Proc. CVPR, 2016, pp. 770-778.

[8] A. G. Howard et al., ”MobileNets: Efficient convolutional neural networks for mobile vision applications,” arXiv:1704.04861, 2017.

[9] K. Simonyan and A. Zisserman, ”Very deep convolutional networks for large-scale image recognition,” in Proc. ICLR, 2015.

[10] R. R. Selvaraju et al., ”Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in Proc. ICCV, 2017, pp. 618-626. [11] T. Ferentinos, ”Deep learning models for plant disease detection and diagnosis,” Computers and Electronics in Agriculture, vol. 145, 2018.

[12] J. G. A. Barbedo, ”Impact of dataset size and variety on the effectiveness of deep learning for plant disease classification,” Computers and Electronics in Agriculture, vol. 153, 2018.

[13] P. Jiang et al., ”Real-time detection of apple leaf diseases using deep learning approach based on improved convolutional neural networks,” IEEE Access, vol. 7, pp. 59069-59080, 2019.